AISR - Project Instructions, Report Format and Evaluation

CSAI4014 Algorithms for Intelligent Systems and Robotics

B.Tech CSE (AI & ML), Semester VII

Batch: B1 - B2 Date: 14/10/2026

Submission deadline: 29 October 2026

Your group, members and project title are given in the separate file *AISR_Project_Distribution_B1-B2_14-10-2026.xlsx*. This document explains how to do the project, how to prepare and print the report (see also the sample report *AISR_Sample_Project_Report.pdf*), how to prepare the slides and video, how you will be evaluated, and how to submit.

1. Project Instructions

Purpose of this project

This project is designed to build skills you will need in industry:

- Self-learning: study a real problem, find sources, learn new tools (Webots, CoppeliaSim, PyBullet, Python simulation libraries, Git) on your own.
- Group collaboration: plan, divide and track work like an engineering team; hold regular meetings and keep a meeting log.
- Co-learning: teach each other what you learn; every member must be able to explain the whole project, not only his or her part.
- Co-creating: build one working system together, with each member owning a clear technical module.
- Industry readiness: work with version control, written documentation, deadlines, recorded online meetings and peer review, as done in companies.

Steps to follow

Step	What to do
1	Find your group and project in the Excel file (sheet 'Group Allocation') and read the full project description in the sheet 'Project List'.
2	Hold a first team meeting (online or in person). Choose a team lead, divide the work into technical modules, and record decisions in the meeting log.
3	Create one GitHub repository for the group. Every member commits his or her own code from his or her own account.
4	Study the related AISR syllabus topics and at least 5 sources. Agree on the novelty of your work.
5	Build and test the simulation. Collect results (plots, tables) and compare at least two methods or settings.
6	Type the report in the format of the sample report (Section 2 of this document). Divide the writing equally. Fill the Contribution and Writing Allocation tables.
7	Prepare the slides (maximum 17, including title and end slide) as per Section 3 of this document.
8	Record a 20-minute online team meeting (Google Meet / Microsoft Teams / Zoom) where every member is visible and presents his or her part, including a live simulation demo.
9	Fill the peer-assessment sheet (each member, individually) and attach it to the report.
10	Upload and submit as below before the deadline. Be ready for a short viva in class after submission.

What to submit (one submission per group)

1. Printed and bound hard copy of the report (submitted to the course faculty). 2. Report PDF. 3. Slides (one PDF, maximum 17 slides). 4. GitHub repository link. 5. YouTube video link.

2. Report Format

A. General rules for the project report

1. The report must be TYPED and PRINTED (not handwritten). A printed hard copy of the report is MANDATORY and must be submitted to the course faculty, along with the same report as one PDF file through the Google Form.
2. Follow the sample report exactly (AISR_Sample_Project_Report.pdf). You may type the report in the Word template (AISR_Sample_Project_Report.docx) or in Overleaf (upload AISR_Sample_Project_Report_Overleaf.zip). Every page must be properly formatted as given below.
3. Paper and margins: A4 size, portrait. Margins: left 3.0 cm (for binding), right, top and bottom 2.5 cm.
4. Page border: a single-line border on EVERY page, including the title page.
5. Font: Times New Roman. Body text 12 pt, line spacing 1.5, justified. Section headings 14 pt bold, sub-section headings 12 pt bold. Report title on the title page 18-20 pt bold.
6. Page numbers: on every page except the title page, at the bottom centre. Front pages (Declaration, Abstract, Index, Lists) in Roman numerals (i, ii, iii); from Section 1 onwards in Arabic numerals (1, 2, 3).
7. Index (Table of Contents) is MANDATORY, with section names and page numbers. A List of Figures and a List of Tables must follow the Index.
8. Figures and tables must be numbered (Figure 1, Table 1) and have a caption. Every figure and table must be referred to in the text.
9. References are MANDATORY: at least 8 references in IEEE style, numbered [1], [2], ... in the order they are cited, and every reference must be cited in the text.
10. Length: 25 to 35 pages including all front pages. Print single-sided, spiral or soft bound.
11. The writing work must be divided EQUALLY among all members. Fill the Writing Allocation table (Section D) and put it in the report after the Index.
12. Sections 4 (Novelty) and 10 (Contribution) are MANDATORY. A report without them will not be evaluated.

A1. Common title page

1. The title page is COMMON for all groups. Use exactly the title page given in the sample report (also given separately as AISR_Sample_Title_Page.pdf and in the Word file).
2. Change only the project title, Project No., Group No., Batch and member names with SAP IDs. Keep everything else (logo, wording, layout, border) the same.
3. Do not add any other logos, colours or decorations to the title page.

B. Structure of the report (follow this order)

Section	Heading	What to write	Pages (approx.)
Cover	Title page (common format)	The common title page from the sample report: project title, Project No., Group No., Batch, member names with SAP IDs, course name and code, faculty name, month and year.	1
-	Declaration	Statement that the work is original, signed by all members.	1
-	Index (Table of Contents)	All sections and sub-sections with page numbers. Mandatory.	1

Section	Heading	What to write	Pages (approx.)
-	List of Figures / List of Tables	Figure and table numbers, captions and page numbers.	1
-	Writing Allocation	Table D below: who wrote which section.	1
-	Abstract	150-200 words: problem, method, simulation platform, main result, novelty.	1
1	Introduction	Background of the problem, why it matters in real life and in industry.	2
2	Problem Statement and Objectives	Clear problem definition, 3-5 measurable objectives, assumptions and limits.	1-2
3	Literature and Industry Review	At least 5 sources (papers, standards, company products). What exists today and what is missing.	2-3
4	NOVELTY (mandatory)	What is NEW in your work compared with existing solutions or the textbook example: new idea, new combination, new scenario, improved algorithm, better result or new analysis. State it in 3-5 clear points and support each point with evidence from your results.	1-2
5	Syllabus Mapping and Theory	Which AISR syllabus topics are used and the theory behind them (equations, D-H table, transfer function, planning algorithm, etc.).	2-3
6	System Design	Block diagram of the full system, robot/environment model, sensors and actuators, software architecture.	2
7	Methodology and Algorithms	Step-by-step method, flowcharts, pseudo-code of the main algorithms, controller design and tuning.	3-4
8	Simulation Setup	Platform (Webots / CoppeliaSim / PyBullet / Python), world/map description, parameters, how to run the code (commands).	1-2
9	Results and Analysis	Plots and tables (path, error, step response, success rate, time), comparison of methods, discussion of what worked and what did not.	3-4
10	CONTRIBUTION (mandatory)	Technical contribution of EACH member in the table format given below, plus who wrote which section of this report.	1-2
11	Co-learning Reflection	Each member writes 5-8 lines: what I learned on my own, what I learned from team-mates, what I taught others.	1-2
12	Conclusion and Future Scope	Main findings, industrial use of the work, how it could be moved to a real robot.	1
13	References	Mandatory. At least 8 references in IEEE style, all cited in the text as [1], [2], ...	1-2
Appendix	Links and Logs	GitHub repository link, YouTube video link, meeting log (date, duration, attendees, decisions), peer-assessment sheet.	1-2

C. Contribution section format (Section 10, mandatory)

S. No.	Member name and SAP ID	Technical contribution (modules / algorithms / code files / experiments done)	Report sections written + signature
1		e.g. Built the simulation scene and robot model; wrote controller.py (PID heading loop); tuned gains; produced Fig. 6-8.	e.g. Sections 6, 8
2			
3			
4			
5			

Contributions must be specific and technical. Lines like "helped in the project'' or "did research'' are not accepted. Each claim must be traceable to code commits in the GitHub repository or to a part of the video. Each member signs his or her own row in the printed copy.

D. Writing allocation (put this table in the report after the Index)

Member	Name and SAP ID	Sections written by this member	No. of pages written
M1		e.g. Abstract, 1, 2, 13	
M2		e.g. 3, 4	
M3		e.g. 5, 6	
M4		e.g. 7, 8	
M5		e.g. 9, 12	
All	Every member	Section 10 (own row and signature), Section 11 (own reflection), Declaration	-

3. Presentation Slides and Demonstration Video

A. Slide deck rules

1. MAXIMUM 17 slides in total, INCLUDING the title slide and the end (thank you / Q&A) slide.
2. Keep slides simple: few words, large fonts (heading 28-32 pt, body at least 20 pt), one idea per slide, at least one figure or plot on most slides.
3. Use your own diagrams and your own simulation screenshots. Copied figures must carry the source below them.
4. Put the presenter's name in the corner of each slide he or she presents.
5. Tools: PowerPoint, Google Slides or Canva. Submit the slides as one PDF along with the report.

B. Suggested slide plan (17 slides)

Slide	Title	Content	Presenter
1	Title Slide	Project title, Project No., Group No., Batch, member names and SAP IDs, course, faculty.	M1
2	Problem and Motivation	Real-world problem, where it occurs in industry, why it matters.	M1
3	Objectives	3-5 measurable objectives of the project.	M1
4	Existing Solutions	What industry and research already do; the gap.	M2

Slide	Title	Content	Presenter
5	Novelty	What is new in your work, in 3-5 points.	M2
6	Syllabus Mapping	AISR syllabus topics used and where each is used.	M2
7	System Architecture	Block diagram: sensors, controller, planner, actuators, communication.	M3
8	Robot and Environment Model	Robot type, kinematics / dynamics model, world or map.	M3
9	Algorithm 1	Main algorithm (planning / control / perception) with flowchart.	M3
10	Algorithm 2 / Controller Design	Second algorithm or controller tuning and parameters.	M4
11	Simulation Setup	Platform, tools, parameters, how to run (commands).	M4
12	Live Demonstration	Screen-share the running simulation (switch to the simulator here).	M4
13	Results 1	Main plots: path, error, response curves.	M5
14	Results 2 and Comparison	Table comparing methods or settings; discussion.	M5
15	Team Contribution	Who did what (technical), as in report Section 10.	M5
16	Conclusion and Industry Use	Key findings, industrial use, future scope, what we learned as a team.	All
17	End Slide	Thank you, GitHub link, contact e-mails, Q&A.	All

For a group of 4, share the M5 slides between members. The presenter split above is a suggestion; every member must present about the same time.

C. Demonstration video rules

1. Record the presentation as an ONLINE TEAM MEETING on Google Meet, Microsoft Teams or Zoom, and record the WHOLE meeting.
2. Every member must join from his or her own device with CAMERA ON and be VISIBLE for the entire recording (use gallery / grid view).
3. Duration: 20 minutes (accepted range 18-22 minutes). Suggested split: about 14 minutes slides + 6 minutes live simulation demo.
4. Every member must speak and present his or her own part (about 4 minutes each). The live demo must show the simulation actually running, not only screenshots.
5. Start the video with each member saying his or her name and SAP ID on camera.
6. Do not edit out or cut parts of the meeting. Small trims at the start and end are allowed.
7. Upload to YouTube as described in Section 5 of this document.

4. Rubrics and Evaluation Criteria

The project is evaluated out of 100 marks. Part A gives the same marks to all members; Part B is marked separately for each member.

Part A - Group assessment (60 marks)

No.	Criterion	Marks	Type	Excellent (90-100%)	Good (70-89%)	Satisfactory (50-69%)	Needs improvement (<50%)
A1	Problem understanding and syllabus linkage	10	Group	Clear real-world problem; correct and deep use of AISR concepts.	Clear problem; most concepts used correctly.	Problem stated; weak link to syllabus.	Unclear problem; little or wrong theory.
A2	Novelty and industry relevance	10	Group	Clear, original novelty backed by results; strong industrial use case.	Some new element; industrial use explained.	Minor change to a standard example.	No novelty section or copied work.
A3	Technical implementation (simulation works)	15	Group	Complete, robust simulation; clean, documented code on GitHub.	Works with small issues; code mostly clear.	Partly working; limited features.	Not working or not shown.
A4	Results, analysis and validation	10	Group	Quantitative plots/tables, comparisons, honest discussion of limits.	Good plots, some comparison.	Few results, little analysis.	No real results.
A5	Printed report (format, clarity, equal writing)	10	Group	Follows the sample format fully: border, index, page numbers, references; writing shared equally.	Minor gaps in format or balance.	Several sections weak or writing unequal.	Format not followed / mandatory sections missing.
A6	Slides (max 17) and video quality	5	Group	Clean slides, within limits, timing 18-22 min, all visible.	Small lapses in design or timing.	Crowded slides or timing off.	Rules not followed.
Total A		**60**					

Part B - Individual assessment (40 marks)

No.	Criterion	Marks	Type	Excellent (90-100%)	Good (70-89%)	Satisfactory (50-69%)	Needs improvement (<50%)
B1	Individual technical contribution	15	Individual	Major, clearly traceable module (code commits + report + video).	Clear module, partly traceable.	Small or vague contribution.	No visible technical work.
B2	Presentation of own part in the video	10	Individual	Confident, correct, on time, visible throughout.	Correct with small gaps.	Reads slides; weak explanation.	Absent / not visible / did not speak.
B3	Viva / Q&A (understanding of whole project)	10	Individual	Answers on own part AND team-mates' parts correctly.	Good on own part, fair on others.	Only own part, partly.	Cannot explain.
B4	Collaboration and co-learning (peer assessment + reflection)	5	Individual	Rated highly by peers; strong reflection; helped others learn.	Good team member.	Limited participation.	Did not cooperate.
Total B		**40**					
Total	**Final project score (A + B), out of 100 marks**	**100**					

Penalties

No.	Rule broken	Deduction	Note
P1	Late submission (after 29/10/2026)	10% of project marks per day	Not accepted after 3 days.
P2	Report format not followed (no border, index, page numbers or references)	-5 marks from A5	
P2a	Printed hard copy not submitted	Report marks (A5) = 0	
P3	Novelty or Contribution section missing	Report not evaluated	
P4	More than 17 slides	-2 marks from A6	
P5	Video shorter than 18 or longer than 22 minutes	-2 marks from A6	
P6	Member not visible / not speaking in the video	B2 = 0 for that member	
P7	Copied code or report (plagiarism)	Project marks = 0	As per university rules.

Peer-assessment sheet (each member fills for every team-mate; attach in the Appendix)

No.	Team-mate name and SAP ID	Contribution (1-5)	Reliability / on time (1-5)	Helped others learn (1-5)	Quality of work (1-5)
1					
2					
3					
4					

5. Video Upload, Submission and Deadline

The demonstration video must be uploaded to YouTube in UNLISTED mode (private link, not Public). Do not make it Public. (Do not choose 'Private' either, because a Private video cannot be opened by the evaluator.)

The video link, report PDF, slides PDF and GitHub link must be shared through the Google Form that will be sent to you privately. Only one member (the team lead) submits the form for the group.

DEADLINE: the full project, including the video and all documents, must be completed and submitted by 29 October 2026.